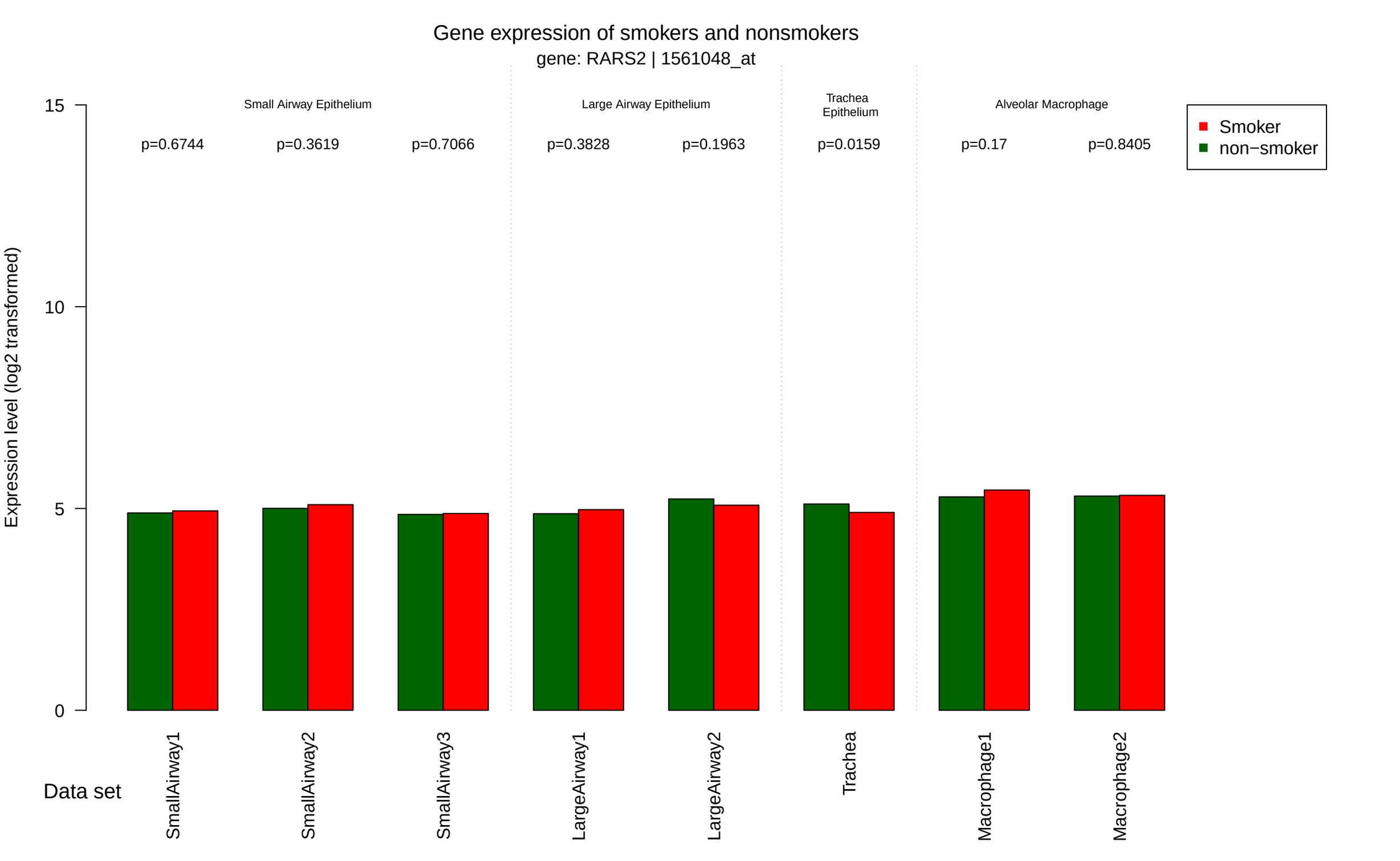
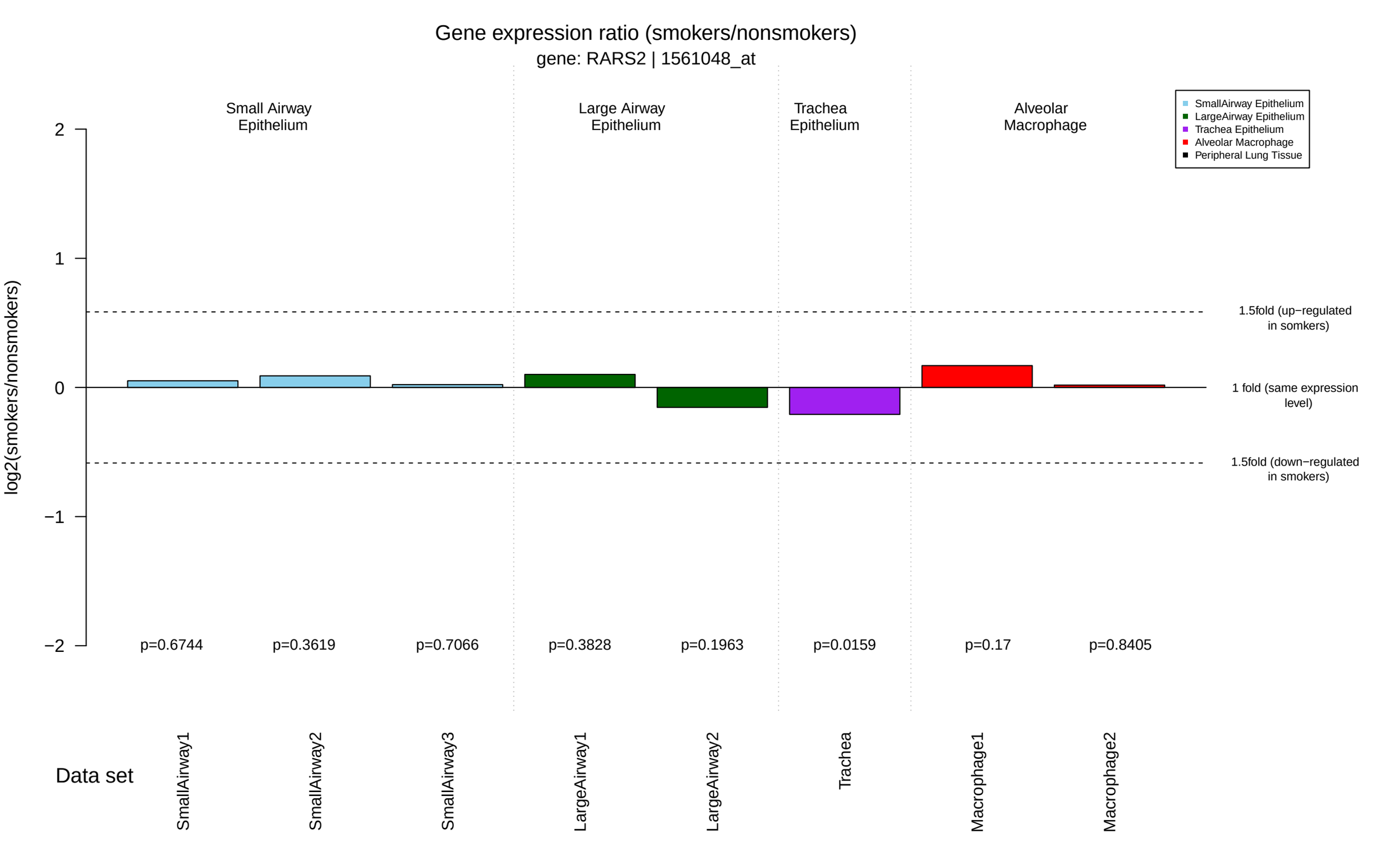
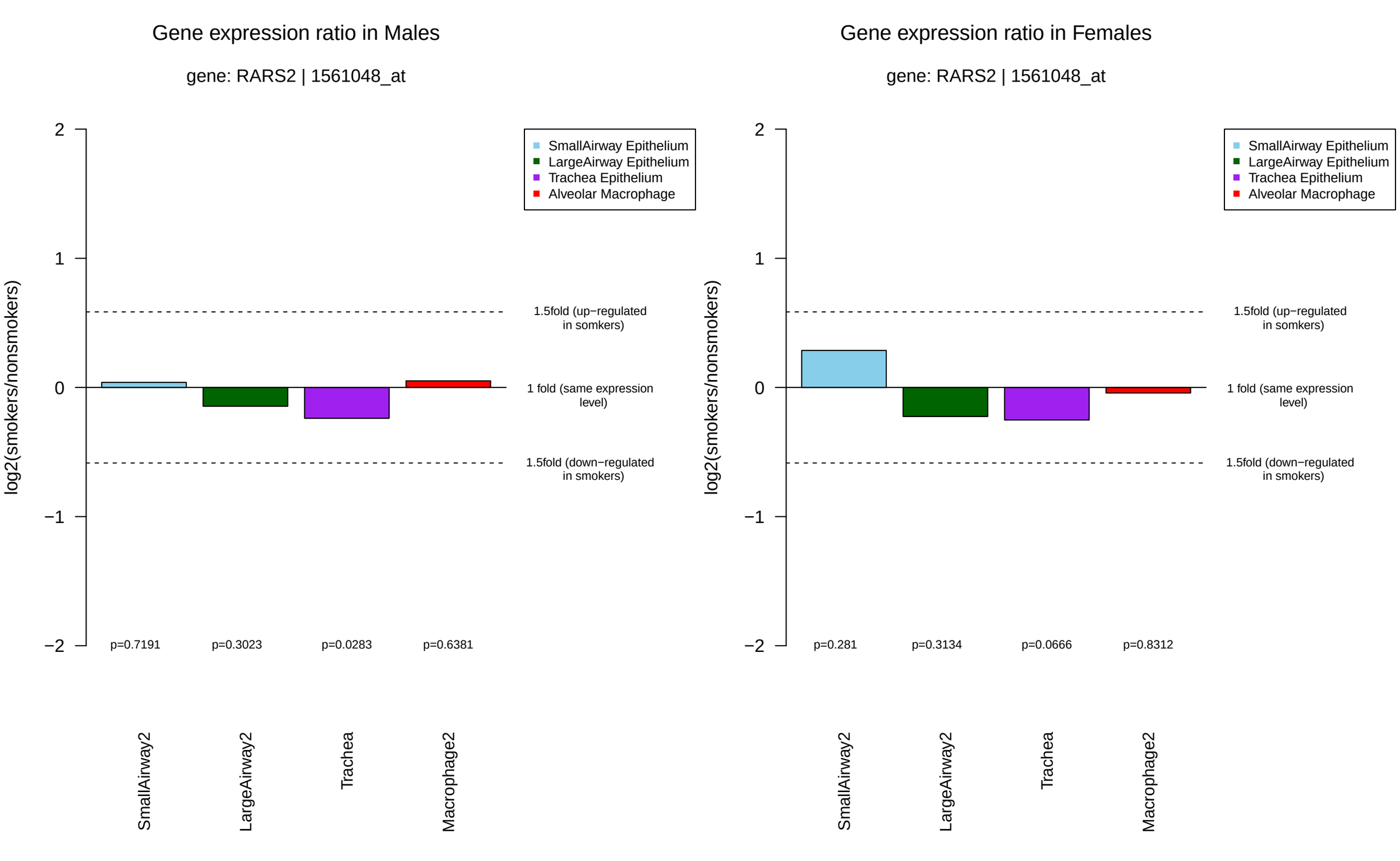
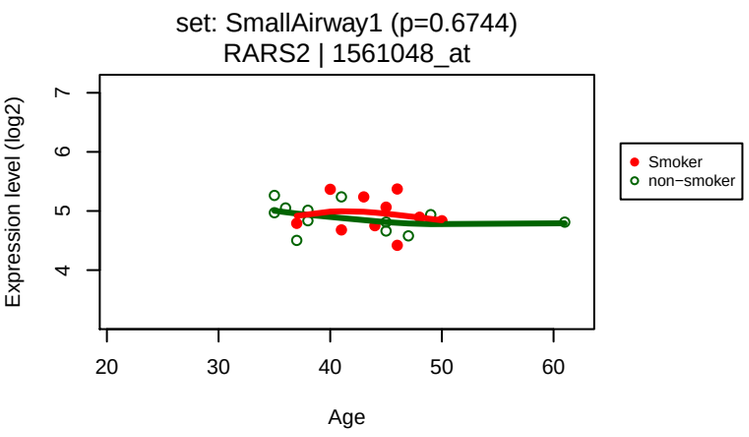
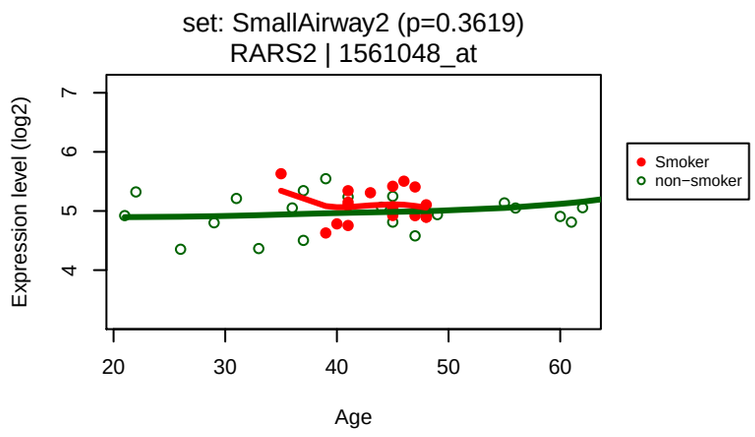
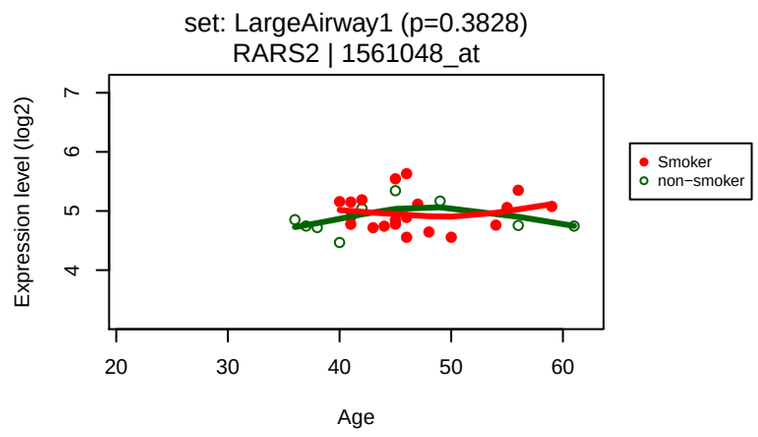

Gene expression of smokers and nonsmokers
gene: RARS2 | 1561048_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6744
p=0.3619
p=0.7066
p=0.3828
p=0.1963
p=0.0159
p=0.17
p=0.8405
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

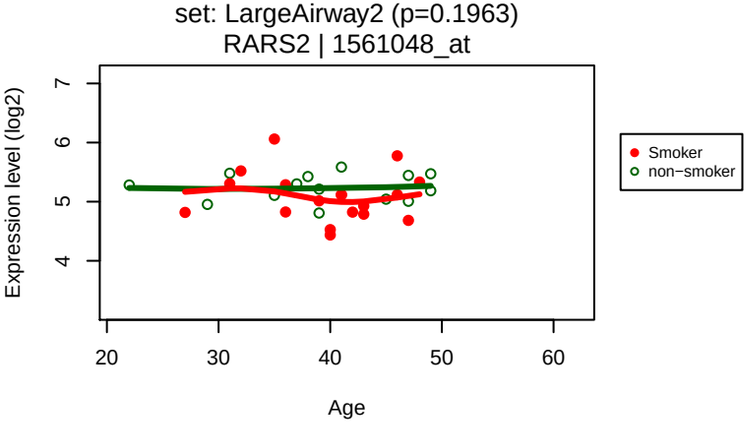

Gene expression ratio (smokers/nonsmokers)
gene: RARS2 | 1561048_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6744
p=0.3619
p=0.7066
p=0.3828
p=0.1963
p=0.0159
p=0.17
p=0.8405
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

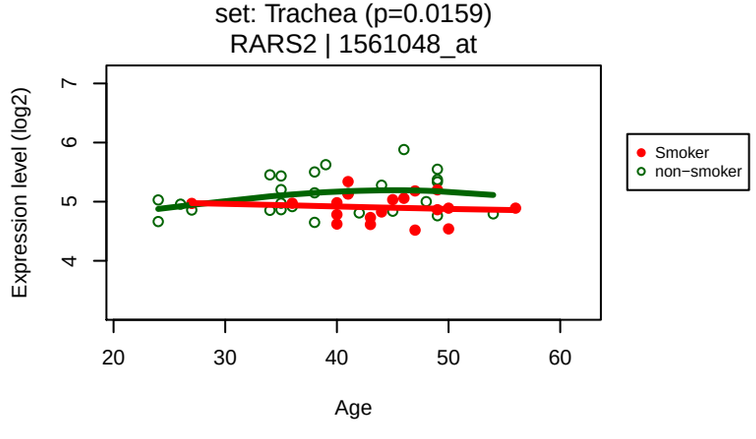

Gene expression ratio in Males
gene: RARS2 | 1561048_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7191
p=0.3023
p=0.0283
p=0.6381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARS2 | 1561048_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.281
p=0.3134
p=0.0666
p=0.8312
SmallAirway2
LargeAirway2
Trachea
Macrophage2

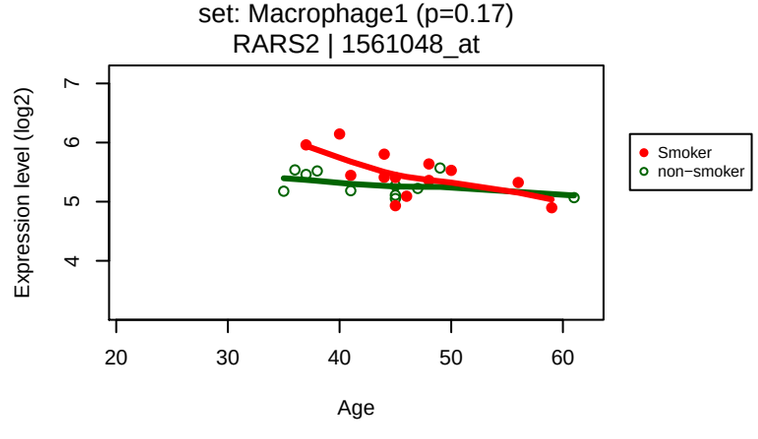

set: SmallAirway1 (p=0.6744)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker

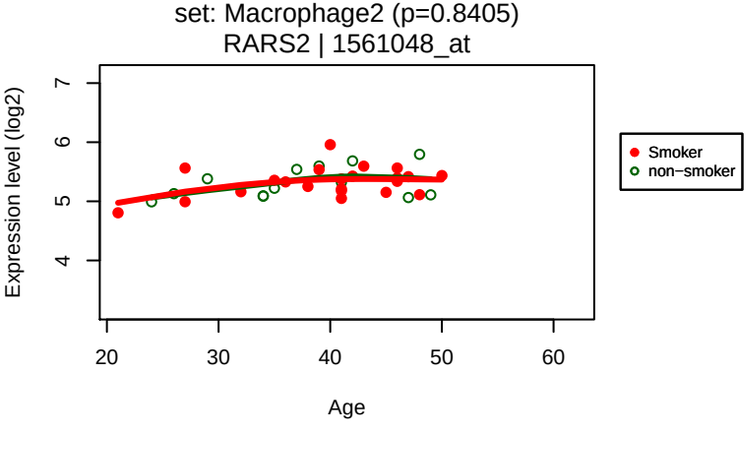

set: SmallAirway2 (p=0.3619)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker

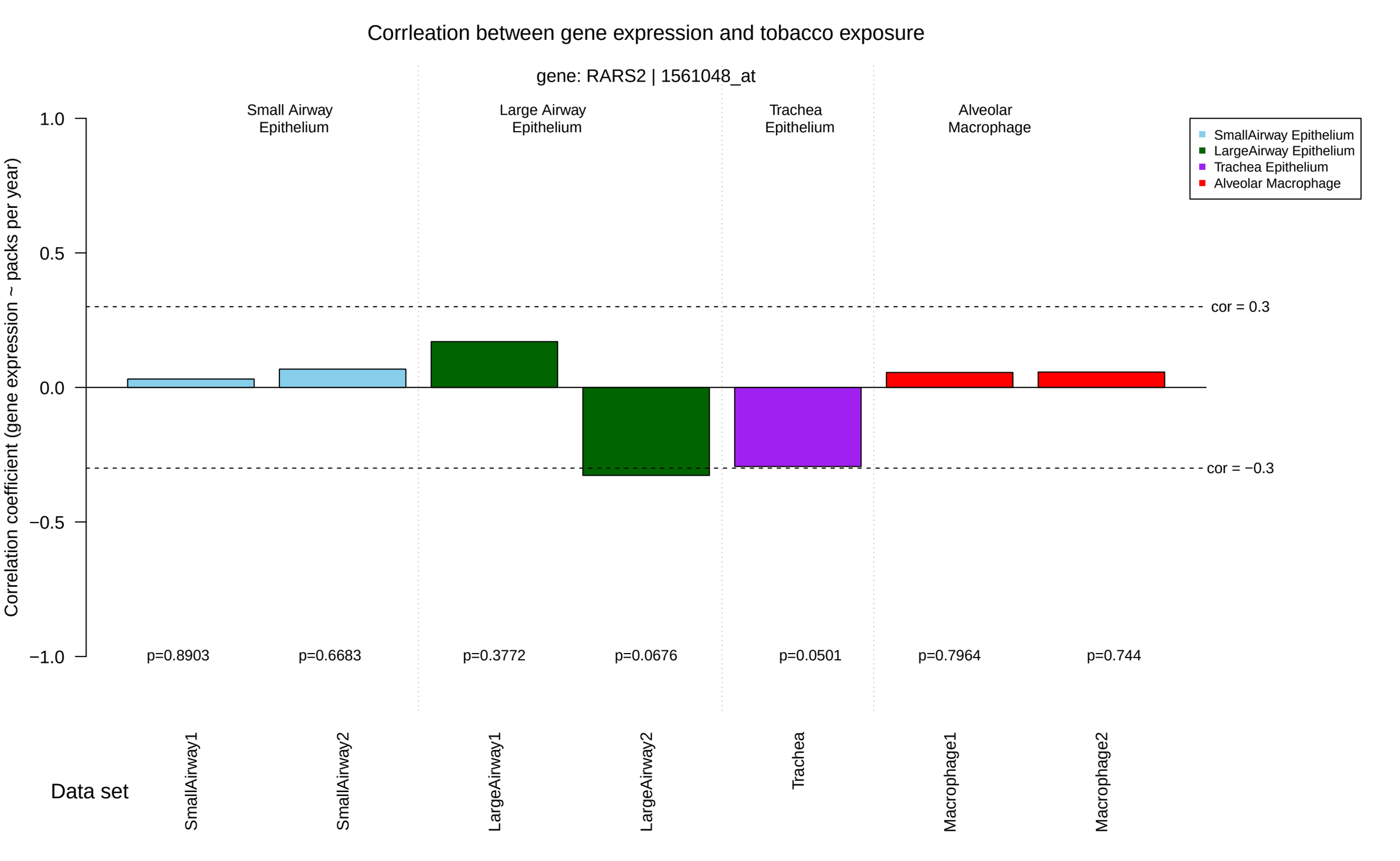

set: LargeAirway1 (p=0.3828)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1963)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0159)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.17)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8405)
RARS2 | 1561048_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARS2 | 1561048_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8903
p=0.6683
p=0.3772
p=0.0676
p=0.0501
p=0.7964
p=0.744
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2